

Macropad schematic and footprints

10/09/2026 14:14:58

<https://claude.ai/chat/47df2b86-bd97-4003-b2aa-19b17e59286c>

Can you check this schematic? I'm making a macropad by wiring a XIAO ESP32C3 to an...

Your schematic matches what you described, and the placement is sensible. I checked...

can you check if this OLED contains i2c pull ups

The datasheet doesn't say. It never mentions pull-up resistors and has no module...

if it already has pull ups and i add the resistors, will that cause a problem

No, it won't cause a problem. Pull-ups on an I2C bus just combine in parallel, and the on...

how do I read this footprint? what are the measurements in mm and whats...

That's a KiCad footprint name, and it's built from a library name, a part type, dimension...

how do I decide on horizontal/vertical alignment?

It comes down to three things: height, board space, and how the part fits your layout....

I'm trying to find the right footprint for this product:...

I'll pull up the product listing to check those dimensions. I couldn't open the Amazon pa...

here's the resistor information: 100pcs 4.7K ohm Resistor 1/4w (0.25 Watt) $\pm 1\%$...

The listing confirms the part is right, but it doesn't give the body dimensions. **What...

why don't they list this information?

There are a few reasons, and none of them are unusual. **The dimension field is for...

do digikey or aliexpress provide measurements/more information for people who are...

DigiKey does, and AliExpress mostly doesn't. **DigiKey** - **Parametric specs.** Resist...

I keep getting power input errors in my schematic, should I make a separate usb-c...

No, you don't need a separate USB-C part. The XIAO already has the USB-C connector o...

so a pwr_flag on 3.3v and gnd on the c3? what about the other components? just 3.3v...

Put the flag on the net, not on a component. One PWR_FLAG on +3.3V and one on GND ...

Can I review this updated wiring with you?

Almost everything looks right, but there's one real mistake: **R1 and R2 are wired in...

like this?

Yes, that's the right topology. Each resistor now hangs off its signal line, with one end o...

so bidirectional on the OLED but input on the XIAO right?

Not quite. Make both ends bidirectional (or passive), not input on the XIAO side. **Why*...

so all these labels should be bidirectional?

Yes, bidirectional works for all six (D2, D3, D7, D8, SDA, SCL), as long as each label has ...

shouldn't I do ouptut for the digital pins coming out of the MCU and then input for the...

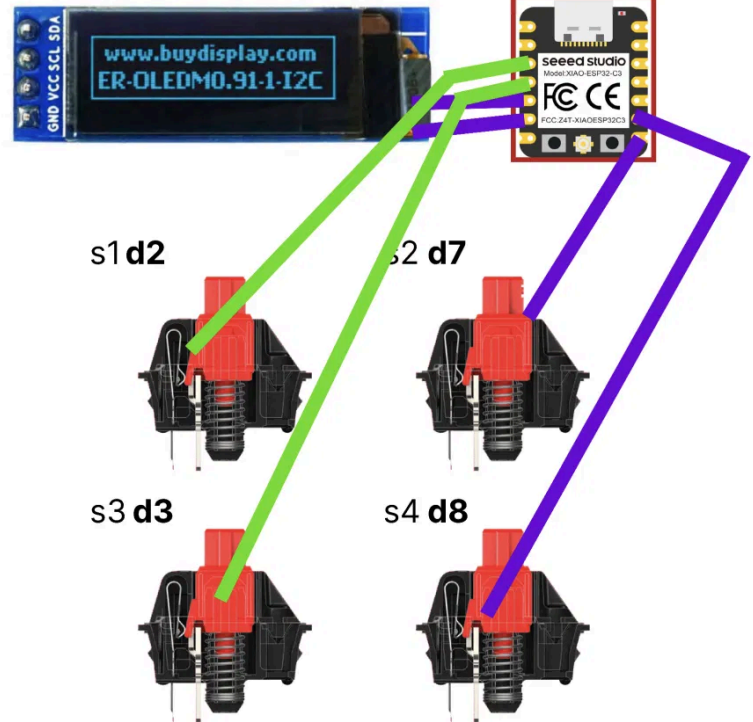
Your instinct is a good convention, but it points the opposite way for these signals, and I...

google says: Microcontroller Input (e.g., Reset line, Interrupt input, Enable pins) Input U...
Not quite. The snippet you found says the opposite of your conclusion, because the sha...
can you verify
Yes, the wiring is correct. I traced every net. ****Verified**** - ****Switches:**** D2 → SW1, D3 ...

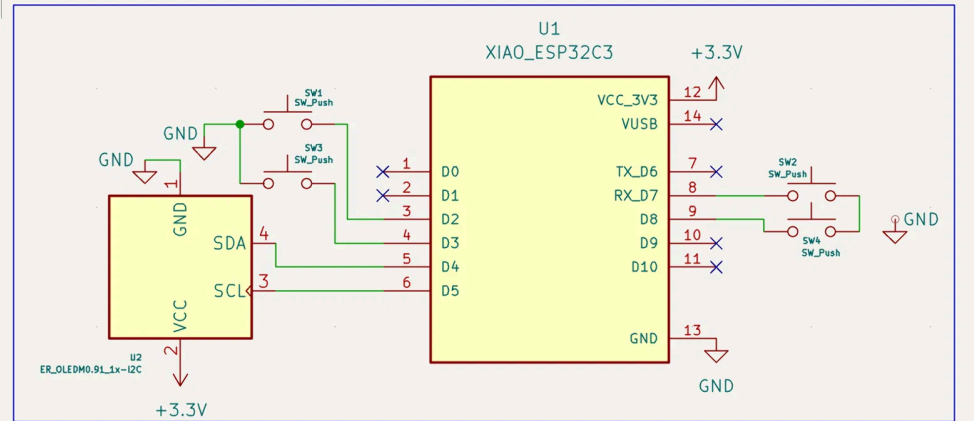


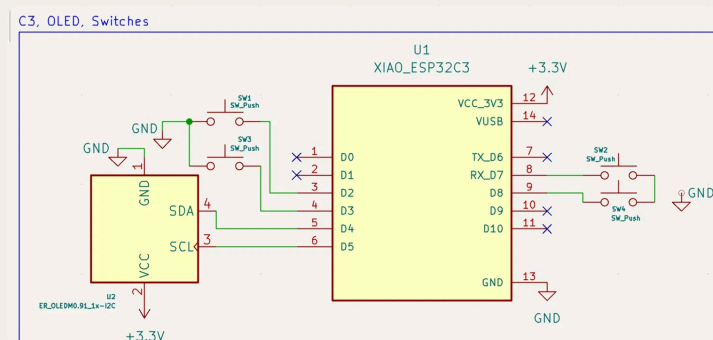
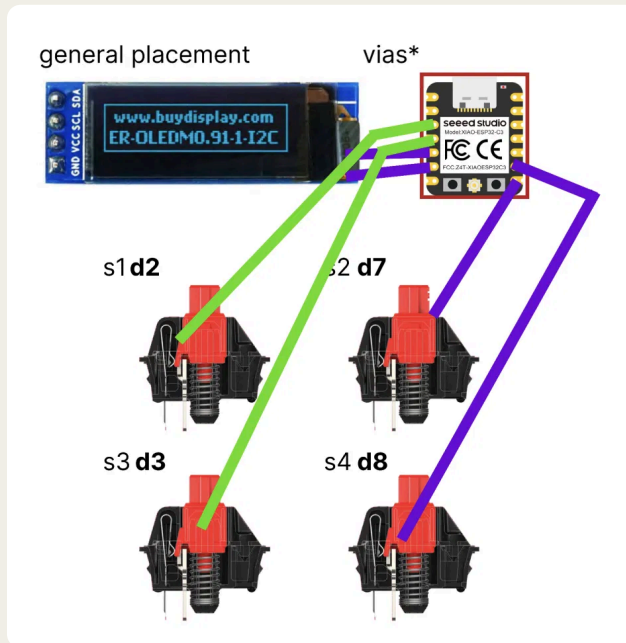
general placement

vias*



C3, OLED, Switches





Can you check this schematic? I'm making a macropad by wiring a XIAO ESP32C3 to an OLED and 4 switches. I want the power to come from the USB-C of the xiao (so I'll design the case to leave it exposed in a smart way). I want it to follow this general placement (unless there's something wrong with it)

To review:

the ER_OLEDMO.91_1x-12c sda pin maps to D4, SCL to D5 AND VCC TO 3.3, SW1 to d2, s3 to d3 and sw2 to d7 and sw4 to d8, the other pins on the xiao (other than gnd and 3.3) have no connect flags. Does this look right?

10/8/2026, 3:15:40 PM



Verifying power pins and module footprint pin order. ▾

⌚ Verifying power pins and module footprint pin order.

⌚ Done

Your schematic matches what you described, and the placement is sensible. I checked every connection.

Connections that check out

- **OLED:** pin 1 (GND) goes to ground, pin 2 (VCC) to +3.3V, pin 4 (SDA) to D4 and pin 3 (SCL) to D5. D4/D5 are also the XIAO's default I2C pins, so stock **Wire** examples will work without remapping.
- **Switches:** SW1 goes to D2, SW3 to D3, SW2 to D7 and SW4 to D8. The other side of each switch goes to GND, which is the right way to wire them for internal pull-ups.
- **Unused pins:** D0, D1, D6, D9, D10 and VUSB all have no-connect flags, and only 3V3 and GND are used. Powering everything from the XIAO's USB-C is fine, and the OLED and switches draw far less than the 3V3 regulator can supply.

Things to fix or double-check

1. **I2C pull-ups.** The schematic has none. Many 0.91" modules have them on board, but check the ER-OLEDM0.91-1-I2C datasheet. If they're missing, add 4.7k resistors from SDA and SCL to 3.3V. It's cheap to add DNP footprints just in case.
2. **Firmware pull-ups.** The switches have no external pull-ups, so use **INPUT_PULLUP** and read them as active-low.
3. **D8 is a strapping pin** (GPIO8 on the C3). It's fine in normal use, but don't hold SW4 while plugging in or resetting, because that can interfere with booting. If you want zero risk, swap SW4 to D9 or D10, though D9 is also tied to the boot button.
4. **KiCad ERC.** You'll probably get "power input not driven" warnings on +3.3V and GND. Adding PWR_FLAG symbols on those nets fixes that.
5. **OLED footprint.** Confirm the footprint's pin order (GND, VCC, SCL, SDA) and spacing against the actual module, since pin orders vary between OLED modules.
6. **Software details.** The I2C address will be 0x3C, or 0x3D if the module has an address jumper.

Placement

- Putting the XIAO's USB-C at the top edge, away from the switches, works well for the case.
- In your sketch the OLED's header pins are on its left end, but the XIAO is on its right, so the SDA/SCL traces have to run under the module. That's also where they cross the D2/D3 traces, which is why you need the vias. If you rotate the OLED 180° so its pins face the XIAO, the I2C traces get short and the crossing probably goes away. The screen would then be upside down, but the flip is one line in firmware (e.g. U8g2's **R2** rotation).
- Your D7/D8 routing is nested, so those traces don't cross each other.
- Each switch's second pin needs a ground connection. A ground pour on the back layer makes that easy.

10/8/2026, 3:16:34 PM



can you check if this OLED contains i2c pull ups

10/8/2026, 3:33:36 PM



Checking the datasheet for pull-up and pin order details. ▾

⌚ Checking the datasheet for pull-up and pin order details.

✅ Done

The datasheet doesn't say. It never mentions pull-up resistors and has no module schematic, only the pinout, outline drawings and electrical specs. I can't confirm either way from this document.

What the datasheet does confirm

- **Pin order:** pin 1 GND, 2 VCC, 3 SCL, 4 SDA, which matches your schematic (U2 pins 1-4).
- **Supply:** VDD is 3.0 to 5.0 V, so 3.3 V from the XIAO is fine.
- **Current:** about 23.5 mA typical and 27 mA max at full-screen-on, which is trivial for the XIAO's 3V3 rail.
- **Header:** 4 pins at 2.54 mm pitch, 7.62 mm end to end, so your footprint should be a plain 1x4 header.
- **Controller:** SSD1306, 128x32. The datasheet gives no I2C address, but SSD1306 modules are normally 0x3C.

Options for the pull-ups

1. **Measure the module.** With it unpowered, put a multimeter between SDA and VCC, then SCL and VCC. A reading around 4.7k to 10k means pull-ups are on board. An open circuit means there are none.
2. **Add pull-up footprints to your PCB.** Put 4.7k resistors from SDA and SCL to 3.3V. If the module already has pull-ups, leave yours unpopulated (DNP). If it does, two sets in parallel is harmless: 4.7k in parallel with 4.7k is about 2.35k, well within spec for a single 0.91" OLED. The XIAO has no I2C pull-ups of its own.
3. **Software fallback.** The ESP32-C3's internal pull-ups are weak, roughly 45k. They often work at 100 kHz on short traces, but they're marginal, so I wouldn't rely on them.

I'd go with option 2 so you're covered regardless of what the module has.

10/8/2026, 3:33:47 PM





if it already has pull ups and i add the resistors, will that cause a problem

10/8/2026, 3:52:05 PM



Checking whether extra resistors conflict with existing pull-ups. ▾

🕒 Checking whether extra resistors conflict with existing pull-ups.

🕒 Done

No, it won't cause a problem. Pull-ups on an I2C bus just combine in parallel, and the only real risk is making the combined resistance too low.

Why it's safe

- Two pull-ups in parallel give a lower total resistance. A 4.7k on your board plus a typical 4.7k on the module comes to about 2.35k, and 4.7k with 10k comes to about 3.2k.
- The limit comes from how much current the devices must sink to pull the line low. The I2C spec assumes 3 mA at 3.3 V, which puts the minimum total resistance around 1k. Anything above that is fine.
- At 2.35k, each line draws about 1.4 mA while held low, which is negligible.
- Lower total resistance also gives faster rising edges, which helps at 400 kHz or on longer traces.

When it could matter

If the module has unusually strong pull-ups, like 1k or 2.2k, adding 4.7k on top pushes the total to roughly 0.8k to 1.5k. That's right at the edge, and it's the only case I'd worry about. Most of these cheap SSD1306 breakouts use 4.7k or 10k, so it's unlikely.

Practical approach

- Put the footprints on your PCB and measure the module first. If it already has pull-ups, leave yours unpopulated (DNP).
- If it has none, populate yours.
- If you measure something under about 2k on the module, leave yours off.

10/8/2026, 3:52:13 PM





16 Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal

16 Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal

how do I read this footprint? what are the measurements in mm and whats horizontal/vertical?

10/9/2026, 12:48:56 PM



Decoding a specific KiCad resistor footprint name. ▾

🕒 Decoding a specific KiCad resistor footprint name.

✅ Done

That's a KiCad footprint name, and it's built from a library name, a part type, dimensions and an orientation.

Breaking it down

- **16**: just the row number in KiCad's footprint list. It isn't part of the name.
- **Resistor_THT:**: the library. THT means through-hole technology (leads through holes, as opposed to SMD).
- **R_Axial**: a resistor with an axial package, meaning one lead out of each end of a cylindrical body, like a standard 1/4 W resistor.
- **DIN0204**: the DIN body size standard. 0204 is the common small size for 1/4 W resistors (roughly 0.4 in x 0.2 in, so 4 x 2 mm nominal).
- **L3.6mm**: body **length**, 3.6 mm. This is the barrel of the resistor, not counting the leads.
- **D1.6mm**: body **diameter**, 1.6 mm. This is how thick the barrel is.
- **P7.62mm**: **pitch**, 7.62 mm (0.3 in). This is the center-to-center distance between the two holes, so it's the number that matters for your board. It's a standard 3 x 2.54 mm grid spacing.
- **Horizontal**: the mounting orientation.

Horizontal vs. vertical

- **Horizontal** means the resistor lies flat on the board, with one lead bent down at each end. The two holes sit 7.62 mm apart and the body spans between them.
- **Vertical** means the resistor stands on end, with one lead running straight down and the other bent back down alongside the body. The holes are much closer together (often 2.54 mm), and the footprint is named something like **..._P2.54mm_Vertical**. It saves board space but sticks up taller.

For your pull-ups



A 1/4 W resistor with 7.62 mm spacing fits this footprint fine. You're also not stuck with it. If you'd rather place them tight, search for **P2.54mm_Vertical**, or use a 0603 or 0805 SMD footprint if you're soldering by hand and want them small. Any of these is fine electrically.

10/9/2026, 12:49:04 PM



how do I decide on horizontal/vertical alignment?

10/9/2026, 1:04:15 PM



Weighing resistor orientation choices for a macropad build. ▾

🕒 Weighing resistor orientation choices for a macropad build.

🕒 Done

It comes down to three things: height, board space, and how the part fits your layout.

Height and clearance

- **Horizontal** parts sit low, about 2 mm tall for a 1/4 W resistor. They're the better choice when something sits close above the board, like a case lid, the OLED, or a plate.
- **Vertical** parts stand about 6 to 7 mm tall. That's only OK if you have headroom.

Board space

- **Horizontal** takes more length, 7.62 mm between holes plus the pad sizes. It's a bad fit in tight areas.
- **Vertical** takes a small footprint (about 2.54 mm between holes), so it's the choice when space is tight but height isn't.

Routing and assembly

- Vertical has the two pads close together, so you can often connect the pull-up to SDA/SCL and 3.3V with short traces.
- Horizontal is a bit easier to solder and sits more securely, since the body lies against the board. A vertical resistor can lean or wobble if you don't solder it flush.

For your macropad

The pull-ups are tiny, and they're only needed if the module has none. I'd do one of these:

1. Use **horizontal** footprints, leave them unpopulated (DNP), and place them somewhere with no height conflicts, like the back side of the board away from the XIAO and OLED.
2. Switch to **0805 or 0603 SMD** footprints. They're the lowest profile and are easy to hand-solder at 0805. They also don't need the pitch decision at all.

If your case is tight at all, choose horizontal or SMD. Vertical only makes sense if board area is your constraint, which on a four-key macropad it probably isn't.

One more thing: the "Horizontal" in the footprint name describes how the part is mounted, not how it's rotated on the PCB. You can still rotate any footprint in 90° steps in the PCB editor (press R) to line

it up with your traces.

10/9/2026, 1:04:26 PM



```
17 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical
18 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P5.08mm_Vertical
19 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
20 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
21 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P15.24mm_Horizontal
```

```
17 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical
18 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P5.08mm_Vertical
19 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
20 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
21 Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P15.24mm_Horizontal
```

I'm trying to find the right footprint for this product:

<https://www.amazon.com/dp/B07HDFHPP3?>

[niid=nl_cl_lst_a_1_1&ref_=nl_cl_lst_a_1_1&nrid=AMSVW953F9Z4BAB6X13X&th](https://www.amazon.com/dp/B07HDFHPP3?niid=nl_cl_lst_a_1_1&ref_=nl_cl_lst_a_1_1&nrid=AMSVW953F9Z4BAB6X13X&th)

I think the dimensions are - please double check

Shell (body) length: 6mm

Max body diameter: 2.5mm

I don't think this is right or I can clip it

Total resistor length: 58mm

Pitch ~ 58mm total/26mm per side

and I want it to be mounted vertically so I've narrowed down to these footprints

but none of them seem to have the right pitch, could you advise? Or could I just cut the resistor's lead to fit it better? Or is pitch negligible?

10/9/2026, 1:13:29 PM



I'll pull up the product listing to check those dimensions.

Matching the footprint choice to standard mounting conventions. ▾

🔗 Couldn't read the page

🌐 B07HDFHPP3 resistor

🕒 Matching the footprint choice to standard mounting conventions.

✅ Done

I couldn't open the Amazon page (it blocks automated access) and a search for the listing found nothing, so I can't verify your dimensions. Going by the numbers you gave, they look plausible, and your confusion is mostly about what "pitch" means.

Your measurements

- A 6 mm body with a 2.5 mm max diameter matches the DIN0207 footprint (L6.3mm, D2.5mm). These are the common 1/4 W through-hole resistors, so that part of the name is right.
- A 58 mm total length with 26 mm leads per side is normal for a loose resistor with long leads ($26 + 6 + 26 = 58$). You don't need to worry about it.
- Please confirm on the listing that it's the value you want (4.7k for your pull-ups). The power rating doesn't matter here.

Pitch isn't a property of the resistor

Pitch is the distance between the two holes on your board, and you choose it. Axial resistors have long, flexible leads, so you bend them to match whatever spacing the footprint has. The 26 mm per side is lead length, not pitch.

Which footprint to use

Pick #17, `R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical`:

- P2.54 mm is the standard vertical spacing and leaves room for a 2.5 mm body. It also lands on the 0.1" grid, so it's easy to route.
- #18 (P5.08 Vertical) works too, but it just spreads the holes out. Use it only if you want more room to bend the lead.
- #19 to #21 are horizontal, so skip them.

To mount it vertically

1. Stand the resistor on end. One lead goes straight down through its hole.
2. Bend the other lead 180° so it runs back down alongside the body, then through the second hole.
3. Solder both leads, then trim them flush on the back with flush cutters. Trimming is safe and normal. Always leave enough lead to bend before you cut, and trim after soldering rather than before.

Height

A vertical 0207 stands about 6.3 mm tall plus the lead, and more than that if the bent-back lead lifts it off the board. Check that against your case clearance, especially near the OLED. If it's too tight, the horizontal #19 (P7.62) lies flat at about 2.5 mm, or you could use an SMD footprint.

Sources

- High value resistors (electronics) (wikipedia.org)
- List of resistors (wikipedia.org)
- 10 ohms resistors (mcmaster.com)
- protosupplies.com (protosupplies.com)
- 22 ohms resistors (mcmaster.com)
- bb resistors (ti.com)
- resistor 330 ohm 5 14w25 pack (protosupplies.com)
- docs.rs-online.com (rs-online.com)

- powersource.danfoss.com (danfoss.com)

10/9/2026, 1:13:50 PM



here's the resistor information:

100pcs 4.7K ohm Resistor 1/4w (0.25 Watt) $\pm 1\%$ Tolerance Metal Film Fixed Resistor, Multiple Values of Resistance Optional

[Visit the EDGELEC Store](#)

- Metal Film Resistor: Excellent Noise Suppression, Moisture-Proof, High Temperature Resistance And Long Working Life.
- $\pm 1\%$ Tolerance range: Maintaining Sccuracy Over Time
- Product: 4.7K ohm 1/4w (0.25watt) -100 pcs
- Aaxial-Lead Resistors: Wide Variety Of Applications: Arduino, Electronic And Electrical Experiments
- Anti-static Bag Sealed Packaging: Easy to preserve and use.

Product information

Technical Details

Product Dimensions 0.35 x 0.31 x 0.2 inches

Item Weight 0.88 ounces

Manufacturer EDGELEC

ASIN B07HDFHPP3

Item model number [E6P010] 4.7K ohm / 0.25Watt / $\pm 1\%$

Date First Available August 28, 2019

Additional Information

Best Sellers Rank

#3,027 in Industrial & Scientific (See Top 100 in Industrial & Scientific)

#3 in Single Fixed Resistors

Feedback

Would you like to tell us about a lower price?

10/9/2026, 1:15:07 PM



Checking a resistor's specifications and physical dimensions. ▾

🕒 Checking a resistor's specifications and physical dimensions.

🕒 Done

The listing confirms the part is right, but it doesn't give the body dimensions.

What checks out

- **4.7k, 1/4 W, $\pm 1\%$ metal film** is a good pull-up for I2C. Tolerance and power rating don't matter here, since the current is around 1 mA.

- **Axial leaded through-hole** matches the `R_Axial` footprints you've been looking at.

What the listing doesn't tell you

- The "0.35 x 0.31 x 0.2 inches" (about 8.9 x 7.9 x 5.1 mm) can't be the resistor, since the body alone is longer than that. It's almost certainly the dimensions of the bag or box.
- There's no body length, diameter, or lead length, so your 6 mm and 2.5 mm figures aren't confirmed by this listing.

Standard 1/4 W metal film resistors are almost always about 6.3 mm long and 2.3 to 2.5 mm in diameter, which is exactly what the DIN0207 L6.3mm D2.5mm footprint is based on. I'm confident about that from how these parts are sized, but it's not from the listing.

What I'd do

- Use #17 (`R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical`) as planned.
- When the resistors arrive, measure one with calipers or a ruler. If the body is within about 0.5 mm of 6.3 x 2.5, the footprint is fine. A small difference won't cause problems, because the footprint only affects the hole spacing and the courtyard (the clearance outline KiCad draws around the part).
- Since these are just the optional pull-ups, you can leave them unpopulated until you've measured the OLED module.

10/9/2026, 1:15:14 PM



why don't they list this information?

10/9/2026, 1:16:53 PM



Explaining why product listings skip certain measurements. ▾

🕒 Explaining why product listings skip certain measurements.

👌 Done

There are a few reasons, and none of them are unusual.

The dimension field is for shipping, not the part. Amazon's "Product Dimensions" and "Item Weight" fields are often filled with package measurements, which is why you got 0.35 x 0.31 x 0.2 inches. The seller isn't being sneaky, the field just wasn't meant to describe a component.

The power rating stands in for size. Axial resistors are sized by wattage, so "1/4 W" tells an experienced buyer roughly what the body looks like (about 6 mm long and 2.5 mm wide). Sellers assume buyers know that, or they don't think about it.

The audience is hobbyists. These listings are aimed at Arduino and breadboard users, who care about value, tolerance and quantity. Few of them are laying out PCB footprints, so nobody writes a drawing for a bag of 100.

It's a generic part. A real manufacturer like Vishay or Yageo publishes a datasheet with a dimensioned drawing and tolerances. A bulk reseller like this one usually doesn't bother, and the exact size can vary slightly between production batches.

Where to look for the real numbers

- **The photo gallery.** Some sellers add a dimensions image, so scroll through all the images.
- **The Q&A section,** or message the seller directly.
- **A comparable datasheet.** Standard 1/4 W metal film resistors like Yageo's MFR-25 or Vishay's MRS25 are specified at about 6.3 mm long and 2.3 to 2.5 mm in diameter, and these should be about the same.
- **Measuring one yourself** once they arrive. That's the most reliable option, and it takes a minute with calipers or a ruler.

10/9/2026, 1:17:00 PM





do digikey or aliexpress provide measurements/more information for people who are using them for pcbs?

10/9/2026, 1:17:45 PM



Explaining what component specifications a parametric search tool provides. ▾

🕒 Explaining what component specifications a parametric search tool provides.

🕒 Done

DigiKey does, and AliExpress mostly doesn't.

DigiKey

- **Parametric specs.** Resistor listings include a "Size / Dimension" field (for example, 0.098" Dia x 0.248" L, or 2.50mm x 6.30mm), a "Lead Spacing" field (for example, 0.400" or 10.16 mm for horizontal mounting), the mounting type, and the package or case.
- **Datasheets.** Each part links to the manufacturer's datasheet, which has a dimensioned drawing with tolerances. That's what you'd use for a footprint.
- **CAD models.** Many parts have a "PCB Symbol, Footprint & 3D Model" link (usually via Ultra Librarian or SnapEDA), so you can often download a ready-made KiCad footprint.
- **Filters.** You can filter by size, power and tolerance, so you can find a part that fits your footprint instead of the other way around.

DigiKey's catalog is made up of real manufacturers' parts, which is why the data is so complete. Mouser works the same way, and LCSC is similar and well suited to small hobby orders. LCSC pairs with EasyEDA footprints and often lists the dimensions too.

AliExpress

- It works like Amazon: the seller writes the listing, so the quality varies a lot.
- Some listings include a dimension drawing in the photos or a spec table, but many have only a value and a pack size.
- Dimensions are rarely guaranteed, and the part may be unbranded or have no datasheet at all, so a measurement from the real part is the only thing you can trust.

Practical approach for your board

For a part you'll design into a PCB, a DigiKey or Mouser part with a datasheet is the safest, and a 4.7k 1/4 W through-hole resistor costs a few cents there. If you'd rather use the EDGELEC ones you already found, searching DigiKey for a standard 1/4 W axial in the same size class and using its drawing is a good stand-in, as long as you measure the real part when it arrives.

10/9/2026, 1:17:52 PM



I keep getting power input errors in my schematic, should I make a separate usb-c receptable part if I plan to get power into the xiao C3 via an external usb-c/someone plugging it in
which powers other components (the OLED and switches)
what power symbols should I use?

10/9/2026, 1:31:40 PM



Diagnosing ERC power-net errors in the USB-C schematic setup. ▾

🕒 Diagnosing ERC power-net errors in the USB-C schematic setup.

✅ Done

No, you don't need a separate USB-C part. The XIAO already has the USB-C connector on its own board, so the USB-C input, the 5V-to-3.3V regulator and the power path are all inside the module. If you added a receptacle to your schematic, KiCad would put a second connector on your PCB.

Why you're getting the errors

KiCad's ERC only knows a power net is "driven" if some pin on it is typed as a power output (or you've placed a PWR_FLAG). Your OLED's VCC/GND pins and the XIAO's pins are power inputs, and nothing in the schematic says power is coming from outside. The USB cable is off-schematic, so ERC can't see it. It's a bookkeeping warning, not a real fault.

Fixes

1. **PWR_FLAG on GND.** Place one from the power symbol library on the GND net. This tells KiCad that ground is supplied from outside the schematic.
2. **For +3.3V, either option works.**
 - Place a PWR_FLAG on the +3.3V net. This is the quick fix.
 - Or edit the XIAO symbol so pin 12 (VCC_3V3) is typed as a power output. This is more accurate, since the XIAO's regulator really does drive that pin. Then you don't need a flag on 3.3V. It only matters if the error says the net isn't driven.

Power symbols to use

- **+3.3V** and **GND**, which is what you already have, so they're correct. Every pin on those nets is tied together by name, so they're all one net.
- **VUSB** can stay as a no-connect. That pin is the raw 5V from USB, and nothing in your design needs it. If you ever used it, you'd tie it to a **+5V** symbol.

Check the exact error text

The wording tells you what's wrong:

- *"Input Power pin not driven by any Output Power pins"* is solved by a PWR_FLAG as above.
- *"Pin not connected"* means a pin is missing a wire or a no-connect flag, so go and look for it.
- *"Power output pins connected together"* means two outputs share a net (not likely in your design).

If you paste the errors, I can tell you which of these each one is.

10/9/2026, 1:31:52 PM



so a pwr_flag on 3.3v and gnd on the c3? what about the other components? just 3.3v and gnd accordingly?

10/9/2026, 1:36:02 PM



Explaining where power flag symbols belong in a schematic. ▾

🕒 Explaining where power flag symbols belong in a schematic.

🕒 Done

Put the flag on the net, not on a component. One PWR_FLAG on +3.3V and one on GND is enough for the whole schematic.

Why one each is enough

Power symbols with the same name (all your **+3.3V** symbols, all your **GND** symbols) are automatically one net in KiCad, even when they're drawn in different places. So the flag on 3.3V covers the OLED's VCC, the XIAO's pin 12 and everything else on that net. The same goes for GND: the OLED, the XIAO and all four switches are already on one ground net.

What to do with the other components

Nothing special. The OLED just needs VCC wired to a **+3.3V** symbol and GND wired to a **GND** symbol, which you already have. The switches only go to GND on one side and a GPIO pin on the other, so they don't need any power flags.

Placing the flags

- From the power symbols (press **P**, then search **PWR_FLAG**), place one anywhere and connect it to the net with a wire. Putting it next to an existing **+3.3V** or **GND** symbol is the tidiest.
- Make sure its pin actually touches a wire or a pin on that net. A flag floating nearby won't count.

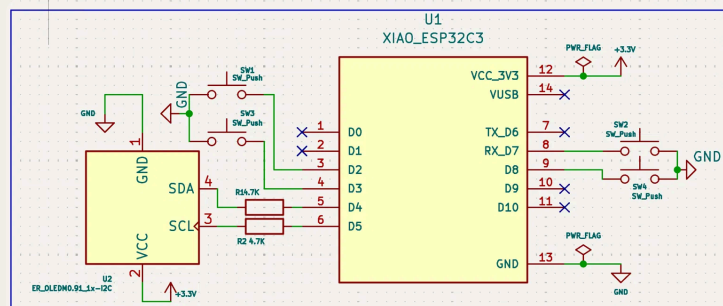
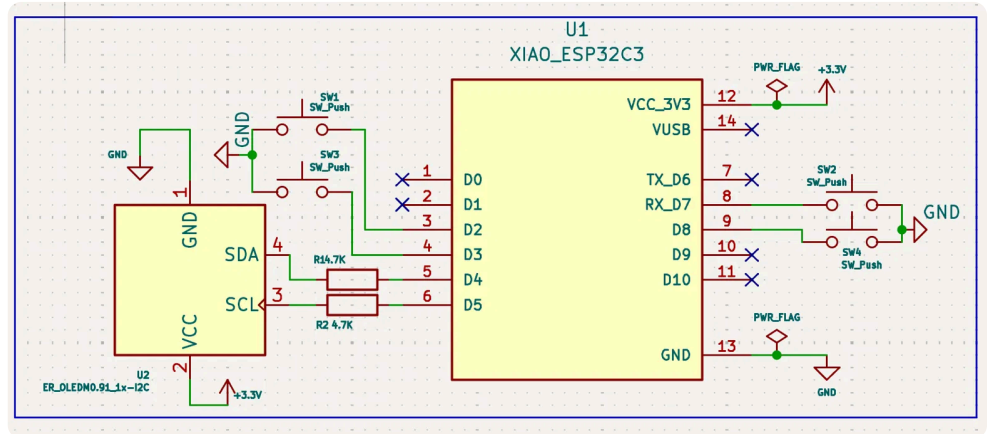
One thing to avoid

Use either a PWR_FLAG on +3.3V or change pin 12 on the XIAO symbol to a power output, not both. A PWR_FLAG is itself a power output, so doing both can give you a "power output pins connected together" error. The simplest route is the flag on both nets and leaving the XIAO symbol alone.

After adding them, run ERC again. If anything still shows up, paste the message and I'll tell you which pin or net it's about.

10/9/2026, 1:36:10 PM





10/9/2026, 1:49:55 PM



- 🕒 Reviewing the resistor placement on the OLED wiring schematic.

Done

The mistake

- ## How to fix it

1. Delete the two resistors from the signal path and reconnect OLED SDA straight to D4 and OLED SCL straight to D5.
2. Rotate R1 and R2 (select, press **R**) so they're vertical.

3. Wire R1's top pin to a **+3.3V** symbol and its bottom pin to the SDA wire between the OLED and D4. Do the same for R2 on the SCL wire. A junction dot should appear where each resistor meets the wire.
4. Keep both resistors, but mark them Do Not Populate in the symbol properties if the module turns out to have its own pull-ups.

What's correct

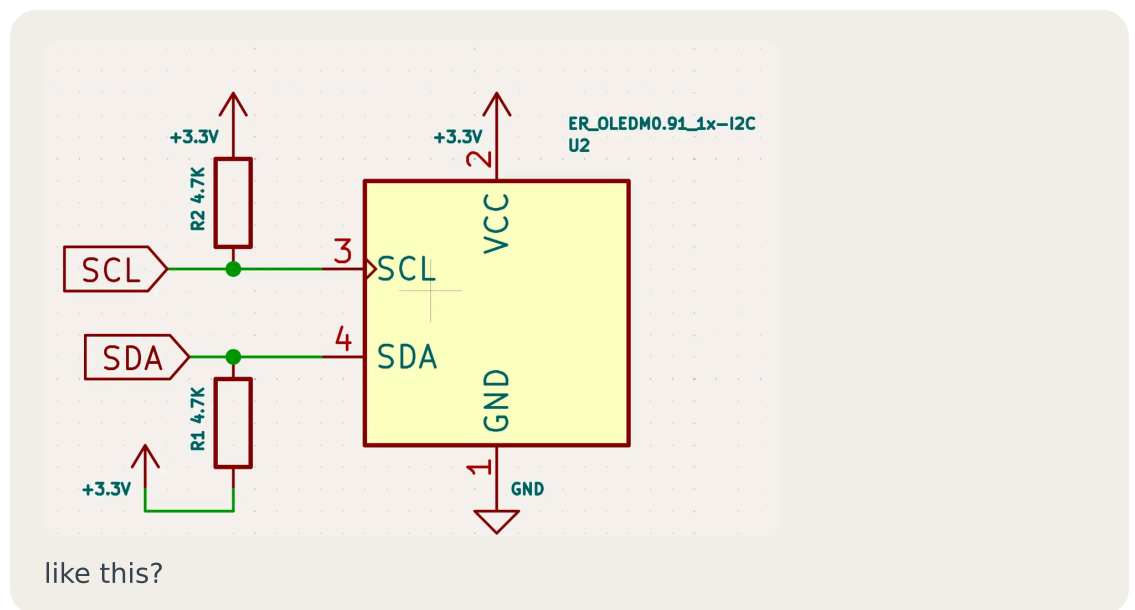
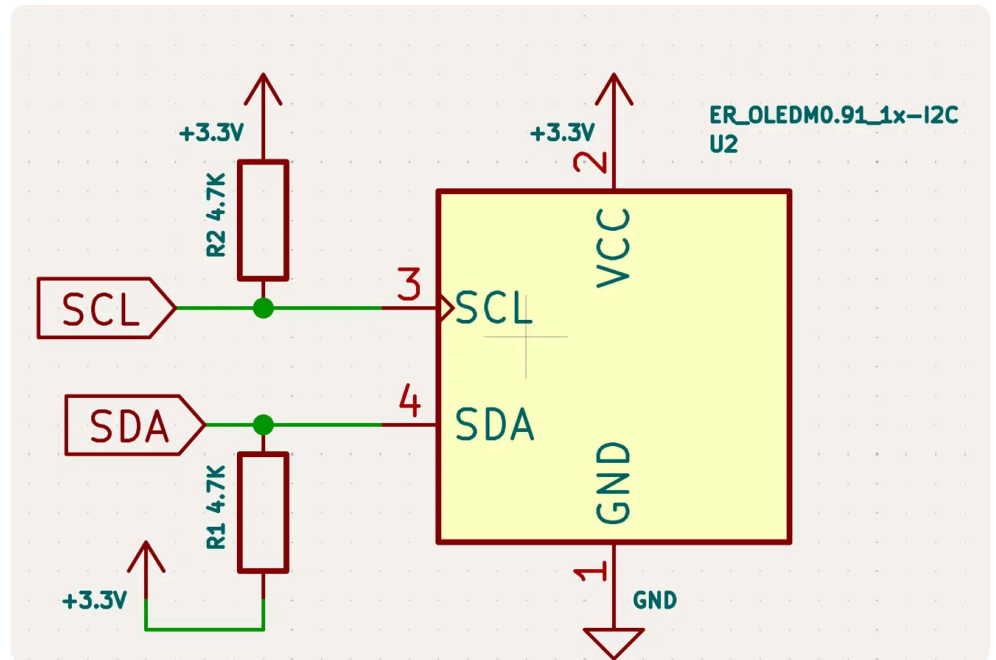
- **Power:** a PWR_FLAG on the +3.3V net at pin 12 and one on GND at pin 13, which is what I suggested. The OLED's VCC goes to +3.3V and its GND to GND.
- **Switches:** SW1 → D2, SW3 → D3, SW2 → D7 and SW4 → D8, with the other side of every switch on GND.
- **Unused pins:** D0, D1, D6, D9, D10 and VUSB all have no-connect flags.
- **I2C pins:** SDA on D4 and SCL on D5 are still right, apart from the resistors.

Small things

- The label "R14.7K" is just the reference R1 and the value 4.7K overlapping. Nudge the text apart so it's readable.
- Check that R1 and R2 have a footprint assigned (the vertical DIN0207 P2.54mm one you chose) before you run the netlist update.

After you move them, rerun ERC. It should come back clean or close to it, and I can look at the new version if you want to check it.

10/9/2026, 1:50:07 PM



10/9/2026, 1:59:24 PM



Verifying pull-up resistor placement and labeling on the circuit. ▾

- ⌚ Verifying pull-up resistor placement and labeling on the circuit.
- ☑ Done

Yes, that's the right topology. Each resistor now hangs off its signal line, with one end on the SDA or SCL wire (junction dots in place) and the other end on +3.3V. The OLED's VCC, GND, SDA and SCL pins all go where they should.

The two things left to check are on the XIAO side:

1. **Labels:** the SCL and SDA labels here have to match labels on the XIAO's D5 and D4 wires. SCL goes to D5 and SDA goes to D4, and the names must match exactly, including case. You can confirm the connection by clicking a label: KiCad highlights everything on that net.
2. **Label type:** if you made these global labels, set the shape to bidirectional or passive. I2C lines carry signals in both directions, and an input or output shape can cause ERC pin-type warnings.

Also, if the labels aren't connected on the other side yet, you'll see "label not connected to anything" warnings, and the OLED won't actually be wired to the XIAO. Before you move on, run ERC again and check that D4 and D5 show up on the same nets as SDA and SCL.

Assign R1 and R2 a footprint (the vertical DIN0207 P2.54mm one you picked). If the OLED module turns out to have its own pull-ups, mark them Do Not Populate.

10/9/2026, 1:59:33 PM



so bidirectional on the OLED but input on the XIAO right?

10/9/2026, 2:00:06 PM



Clarifying pin directionality and label conventions for the connector. ▾

🕒 Clarifying pin directionality and label conventions for the connector.

🕒 Done

Not quite. Make both ends bidirectional (or passive), not input on the XIAO side.

Why

- Label shapes describe the signal direction on the net, not which component the label sits next to. SDA and SCL are one net, so the shape should describe how that net behaves.
- I2C is open-drain and bidirectional. SDA carries data both ways, and SCL can be held low by a device (clock stretching), so neither line is a pure input or output for either chip. An "input" label on the XIAO side could trigger ERC warnings that don't reflect reality.

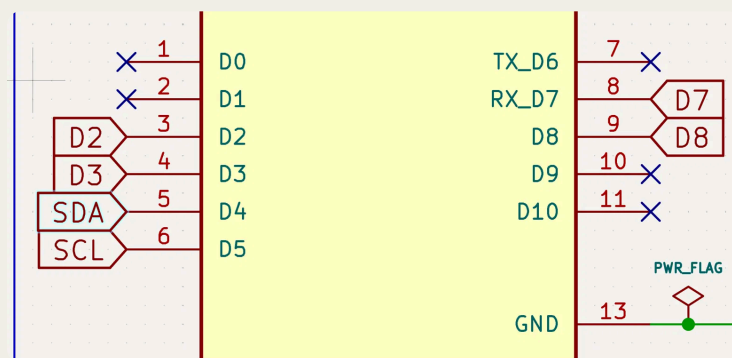
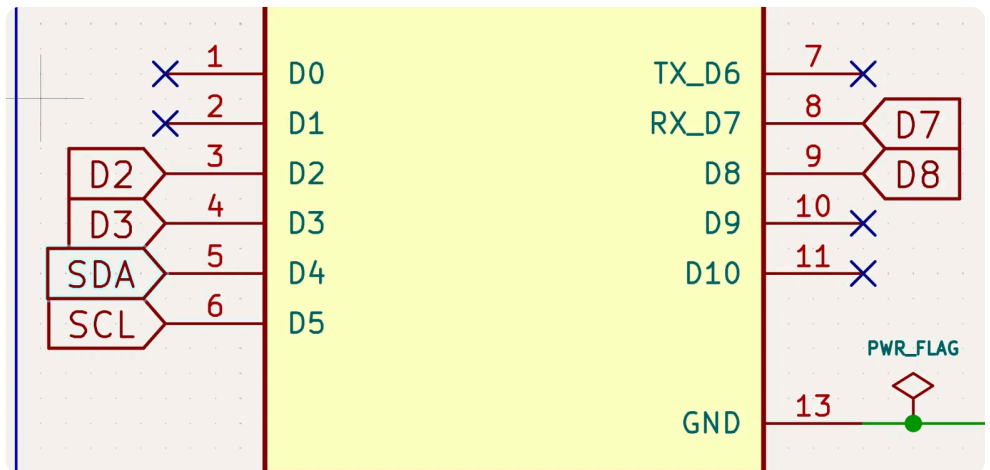
What to do

- Set the SDA and SCL labels to **bidirectional** (or **passive**) on both the OLED and XIAO sides. The shape doesn't need to differ between the two ends.
- Passive is the most neutral if you want the fewest ERC complaints. Bidirectional is the more descriptive choice.

An easier option

Everything is on one schematic sheet, so you don't need global labels at all. A plain **local label** (press **L** instead of **Ctrl+L**) has no shape, no pin-type checks, and works the same way for connecting nets. Delete the arrow-shaped labels and replace them with plain **SDA** and **SCL** labels on each end. Global labels are only worthwhile for multi-sheet designs.

10/9/2026, 2:00:13 PM



so all these labels should be bidirectional?

10/9/2026, 2:01:33 PM



Reading overlapping pin labels in a circuit diagram image. ▾

- 🕒 Reading overlapping pin labels in a circuit diagram image.
- 🕒 Done

Yes, bidirectional works for all six (D2, D3, D7, D8, SDA, SCL), as long as each label has a matching label of the same name on the other end (for example, the D2 label on the switch side). Passive would also be fine. The shape only changes what ERC checks, not what gets connected.

For the switch lines, the real signal direction is a button pressing a pin low, and the XIAO pin is a general-purpose I/O. An input or output shape adds nothing there, so bidirectional or passive avoids false warnings.

These labels are stacked tightly, with D2/D3/SDA/SCL on the left and D7/D8 on the right overlapping each other. Labels overlapping doesn't connect the nets, since KiCad only connects a label at its anchor point on a wire or pin. But it's hard to read and easy to misclick. Two fixes:

- **Switch to plain local labels** (L instead of Ctrl+L). They're text with no arrow shape, so they take up much less room and have no shape setting to worry about.

- **Or add a short wire stub** off each pin and place the labels on the stub ends, spaced out. That also gives you room to click each one individually.

After you change them, click each label and confirm the whole net highlights, including the OLED, resistors or switch on the other end. Then run ERC again.

10/9/2026, 2:01:44 PM



shouldn't I do output for the digital pins coming out of the MCU and then input for the other components that reference those?

10/9/2026, 2:02:37 PM



Clarifying the direction convention for signal labeling. ▾

🕒 Clarifying the direction convention for signal labeling.

🕒 Done

Your instinct is a good convention, but it points the opposite way for these signals, and I also overstated something earlier.

Direction for your signals

- **D2, D3, D7, D8 (switches):** the signal goes from the button to the MCU. The XIAO pins are the inputs (you're reading them with `INPUT_PULLUP`), and the switch is the source. If you wanted directional shapes, the XIAO side would be **input** and the switch side **output**, the reverse of what you described.
- **SDA/SCL:** open-drain and bidirectional, so **bidirectional** (or passive) on both ends, as before.

What I got wrong earlier

I said a wrong shape could cause ERC warnings. In KiCad, the shape on a local or global label is documentation only. ERC doesn't compare it against pin types, so "wrong" shapes won't create errors, and "right" ones won't fix any. The one place it's enforced is hierarchical sheet pins versus hierarchical labels, which you aren't using.

What that means for you

- Shapes only help a human reader follow signal flow. Use them if you like that, with the directions above.
- If you'd rather not think about it, bidirectional or passive everywhere is fine, and so are plain local labels.
- What matters for correctness is that the names match exactly on both ends, so check that the nets highlight end to end.

Your ERC errors will come from pin types and connectivity, not label shapes, so run it again after adding the flags and the pull-ups, and send me anything that remains.

10/9/2026, 2:02:46 PM



google says:

Microcontroller Input

(e.g., Reset line, Interrupt input, Enable pins) Input Use when the digital pin only receives a signal generated elsewhere in the project.

Microcontroller Output

(e.g., Status LED driver, PWM output, Clock out) Output Use when the digital pin drives a signal out to other parts of the project.

so on the XIAO, the labels coming out of the digital pins (not sda or scl) should be output shape and then the other components that use those (like the switches) should be input shapes?

10/9/2026, 2:03:55 PM



Working out which XIAO pins serve as inputs or outputs. ▾

🕒 Working out which XIAO pins serve as inputs or outputs.

🕒 Done

Not quite. The snippet you found says the opposite of your conclusion, because the shape follows which way the signal is flowing, not which part is the MCU.

Applying Google's rule to your pins

- **D2, D3, D7, D8:** the XIAO only receives a signal generated elsewhere (the button press), which is the "Microcontroller Input" row. So the label on the XIAO side is **Input**.
- **The switch side of those nets:** the switch is the source driving the net, so those labels are **Output**.
- **SDA and SCL:** both devices drive and read these lines, so **Bidirectional** on both ends.

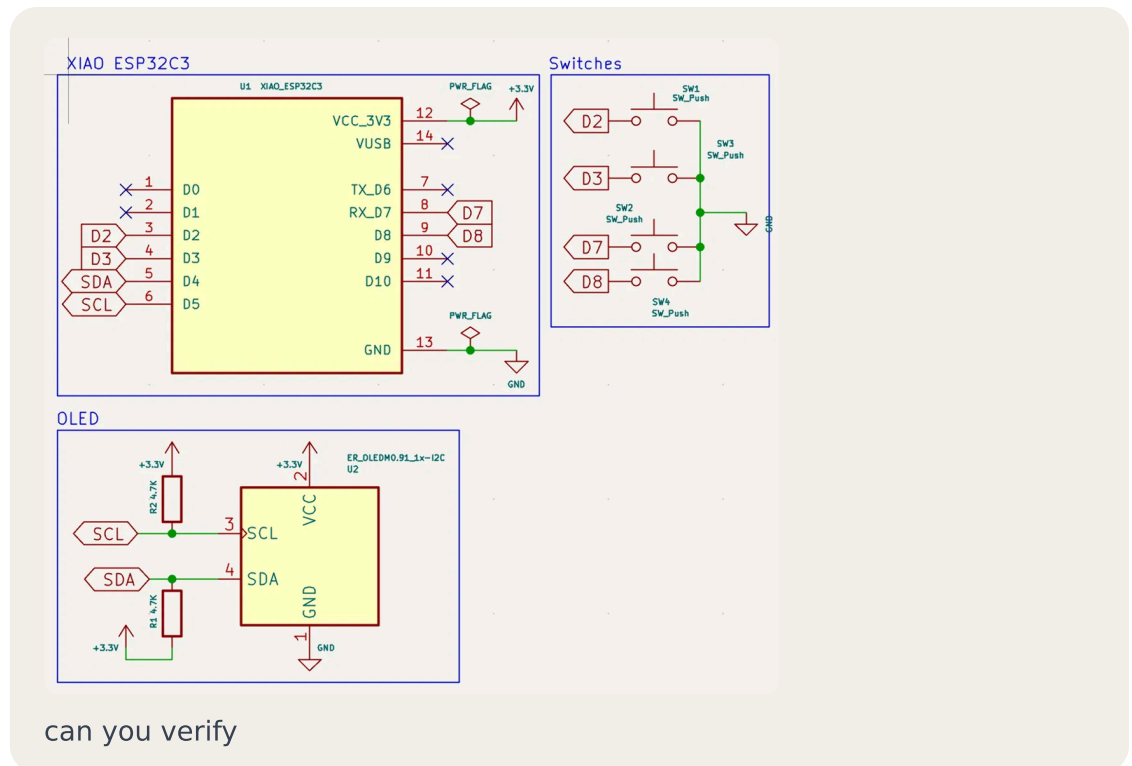
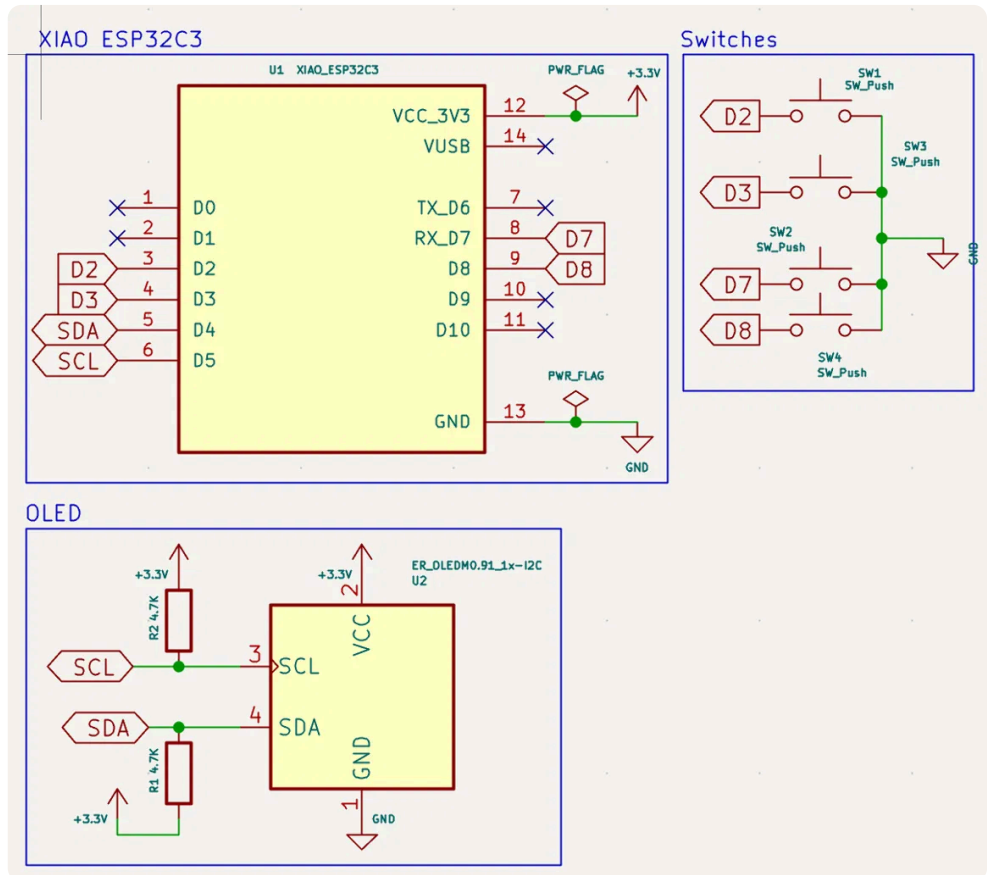
When the XIAO labels would be Output

Only if you configured a pin to drive something, such as an LED, a buzzer, or a chip-select line. You have four buttons and nothing else, so every digital pin on the XIAO is an input.

Reminder

The shapes are only documentation. Unlike hierarchical sheet pins, KiCad won't flag a wrong one, so the connection works either way as long as the names match. Following the convention is a nicety that makes the schematic easier to read, and the nets highlighting end to end is what confirms it's wired correctly.

10/9/2026, 2:04:02 PM



10/9/2026, 2:12:38 PM



Flagging potential ERC conflicts and checking pin assignments. ▾

🕒 Flagging potential ERC conflicts and checking pin assignments.

🕒 Done

Yes, the wiring is correct. I traced every net.

Verified

- **Switches:** D2 → SW1, D3 → SW3, D7 → SW2, D8 → SW4. The label names match the XIAO side, and the right-hand pin of all four switches runs down the shared ground bus, which connects to the GND symbol.
- **I2C:** SDA/SCL labels connect OLED pin 4/3 to XIAO D4/D5. R1 and R2 each sit between +3.3V and their signal line, with junction dots on the wires.
- **OLED power:** VCC (pin 2) goes to +3.3V and GND (pin 1) to GND.
- **Unused pins:** D0, D1, D6 (TX), D9, D10 and VUSB have no-connect flags.
- **Power flags:** one PWR_FLAG on the +3.3V net and one on GND.

Things to watch

1. **ERC on +3.3V.** If it reports "Pins of type Power output and Power output are connected" on the 3.3V net, the XIAO symbol's pin 12 is already typed as a power output. In that case, delete the PWR_FLAG on +3.3V. Keep the GND one.
2. **Label shapes.** D2 and D3 on the XIAO look like plain rectangles, while D7 and D8 have arrow shapes. It's only cosmetic, but pick one style (input for the XIAO side and output for the switch side, as we discussed) so it's consistent.
3. **Reference text.** The SW3 and SW2 reference text sits next to the wrong switches, which makes it hard to read. Move the references so each is clearly beside its own part.
4. **Firmware.** There are no external pull-ups on the switches, so use `INPUT_PULLUP` and read them as active-low.

Next steps

- Run ERC and send me anything that remains.
- Assign footprints:
 - the resistors: `R_Axial_DIN0207_L6.3mm_D2.5mm_P2.54mm_Vertical`
 - the OLED: a 1x04 pin header with 2.54 mm pitch
 - the XIAO: from a Seeed XIAO library
 - the switches: a Cherry MX or Kailh hot-swap footprint, matching the switches you buy
- Then update the PCB from the schematic and start on placement.

10/9/2026, 2:13:00 PM